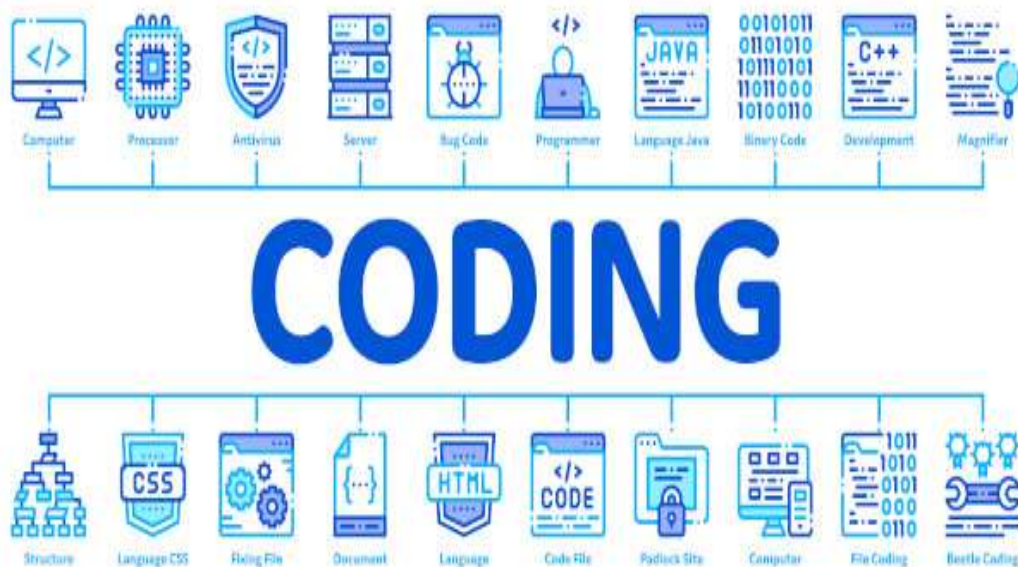


MODULE : OPEN SOURCE CODING

MODULE CODE : OPSC6311

POE – PART 1



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Personal Budgeting Apps Research Report

GoodBudget - Armalene

WalletCorner - Chloe

PocketGuard - Rafiqah

1. Introduction

Personal budgeting applications have become essential tools for individuals seeking to manage their finances effectively in a highly digital and fast-paced world (NerdWallet, 2025). With the widespread adoption of Android smartphones, users increasingly rely on mobile applications to monitor spending, plan budgets, and achieve long-term financial goals (Investopedia, 2025). These applications vary in their design and functionality, ranging from traditional budgeting tools to modern systems incorporating automation and gamification techniques.

Budgeting apps assist users in tracking financial behaviour while also promoting financial literacy and discipline. By providing real-time insights, visual representations of spending, and goal-setting features, these apps enable informed financial decision-making (Forbes Advisor, 2025).

This report examines three Android-based personal budgeting applications: WalletCorner (a gamified budgeting app), Goodbudget (a traditional envelope budgeting system), and PocketGuard (an automated financial tracking app). The purpose of this research is to evaluate each application in terms of its features, strengths, weaknesses, and innovative aspects.

2. Research on Selected Apps

2.1 WalletCorner

Overview

WalletCorner is a gamified personal budgeting application designed to transform financial management into an engaging and interactive experience. Gamification in financial applications has been shown to improve user engagement and consistency in financial tracking (Investopedia, 2025). WalletCorner integrates rewards, achievements, and level progression to motivate users to consistently track expenses and meet financial goals.

Screenshots with Descriptions

Calendar View

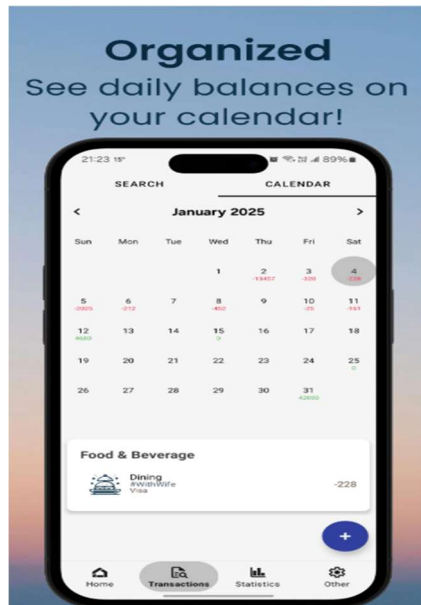


Figure 1: This screenshot shows the calendar feature where daily expenses and income are displayed on specific dates.

Budget Overview

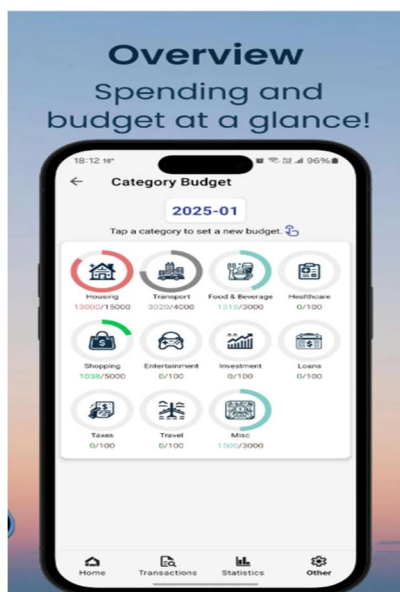


Figure 4: This screenshot shows the budget dashboard, where spending is organised into categories like housing and food. Circular indicators display how much of each budget has been used, helping users stay within limits.

Expense Insights

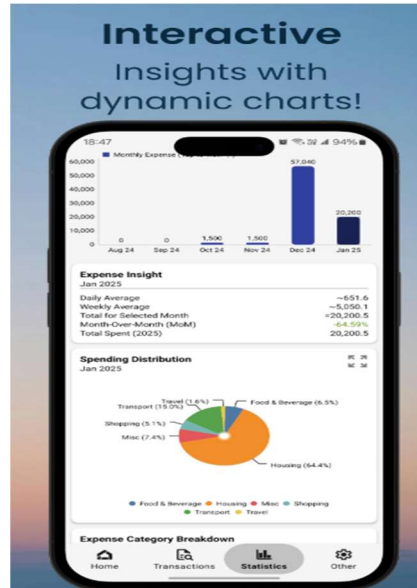


Figure 2: This image presents the analytics section, featuring bar and pie charts that break down spending. It helps users understand trends, compare monthly expenses, and see how money is distributed across categories.

Gamified Feature

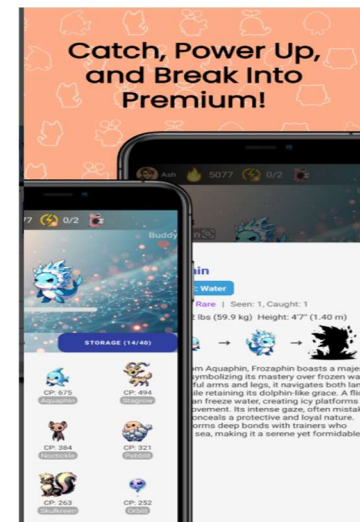


Figure 3: This screenshot highlights the gamification feature, where users collect and interact with characters while tracking expenses. It adds a fun element to budgeting and encourages regular app usage.

Search Function

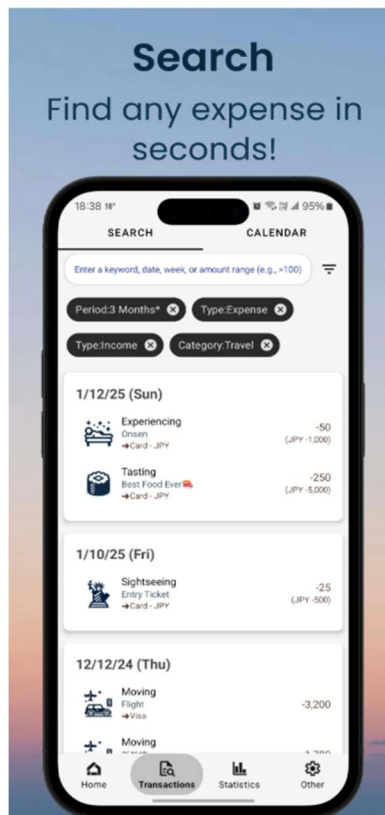


Figure 5: This image presents the search functionality, allowing users to filter and locate transactions by date, category, type, or amount. It makes financial records easy to find and analyse.

Features & Settings

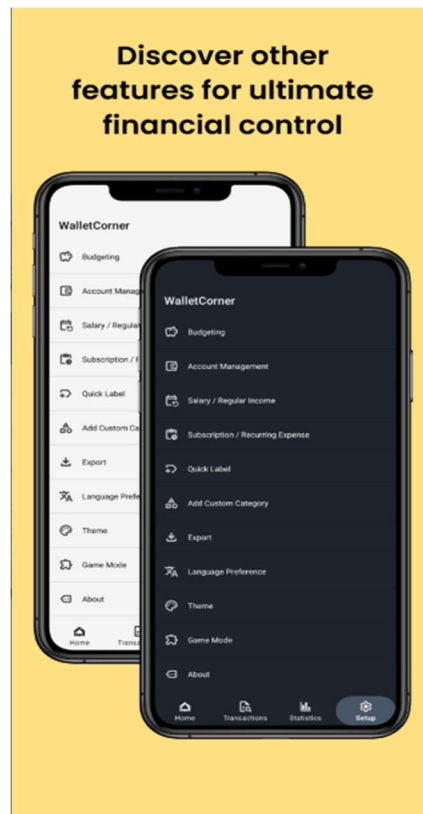


Figure 6: This screenshot displays the features and settings menu, where users can manage budgeting to, accounts, subscriptions, and preferences. It provides full control over how the app is used.

Budget Gamified View & Expense Logging Gamification

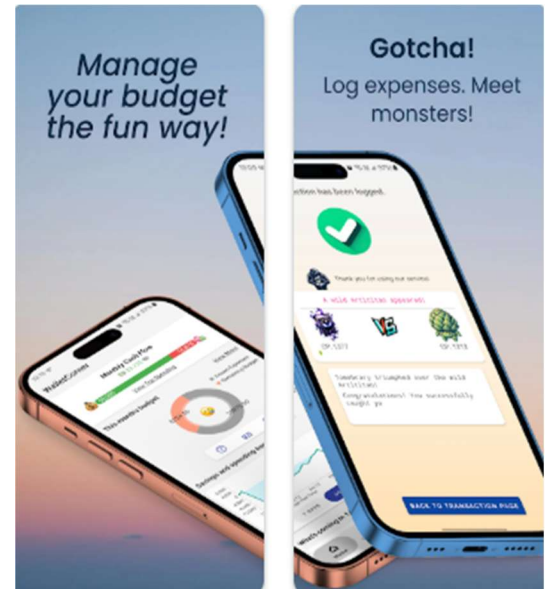


Figure 7: This screenshot shows a more interactive budgeting interface, combining financial tracking with engaging visuals. It makes managing money feel less rigid and more enjoyable.

Strengths

- Highly engaging due to gamification elements, which increase user motivation and consistency
- Encourages regular expense tracking through rewards and achievements
- Provides visual feedback on financial progress, making it easier to understand spending habits

- Makes budgeting less intimidating for beginners by presenting it as a game-like activity

Weaknesses

- May lack advanced financial analysis tools compared to more traditional applications
- Limited availability of academic and professional reviews due to lower popularity
- Gamification elements may distract users from serious financial planning if not balanced
- May not be suitable for users seeking detailed financial reports or investment tracking

Innovative Features

WalletCorner's primary innovation lies in its use of gamification to influence user behaviour. The reward system encourages users to complete financial tasks such as logging expenses or staying within budget limits. As users progress, they unlock levels or achievements, which creates a sense of accomplishment and motivation. This feature is particularly useful because it transforms routine financial management into a more enjoyable activity, increasing long-term engagement. Additionally, the use of visual and interactive elements helps users better understand their financial progress, making complex financial concepts more accessible.

2.2 Goodbudget

Overview

Goodbudget is a widely used envelope-based budgeting system designed to help users control spending through intentional allocation of income. The app mirrors the traditional "cash envelope" method but digitises the process so users can distribute funds across virtual envelopes representing categories such as groceries, transportation, or entertainment (Goodbudget, 2024). The app requires manual data entry for most transactions, which encourages users to remain conscious of their financial habits rather than relying on automated systems.

The interface is clean and minimalist, reducing cognitive load and helping users focus on essential budgeting tasks. It includes features like multi-device sync, shared household budgets, and simple reports that show spending trends and envelope balances.

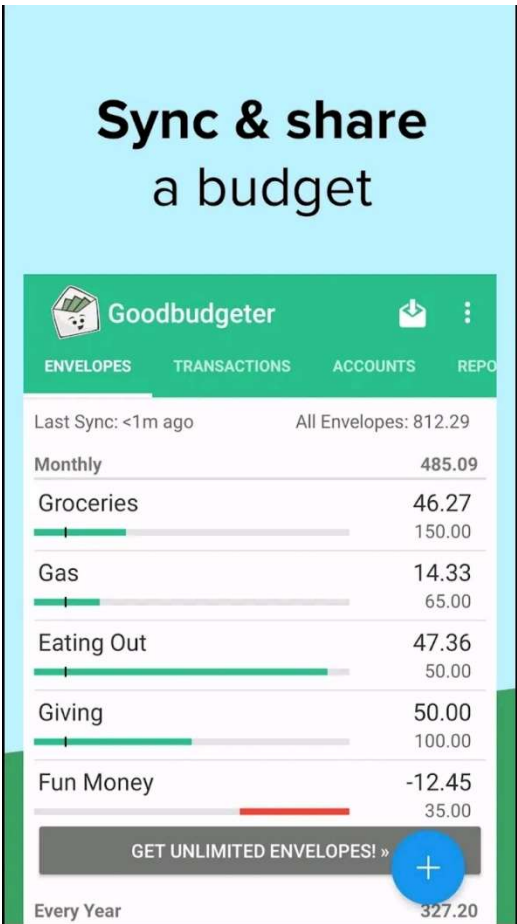
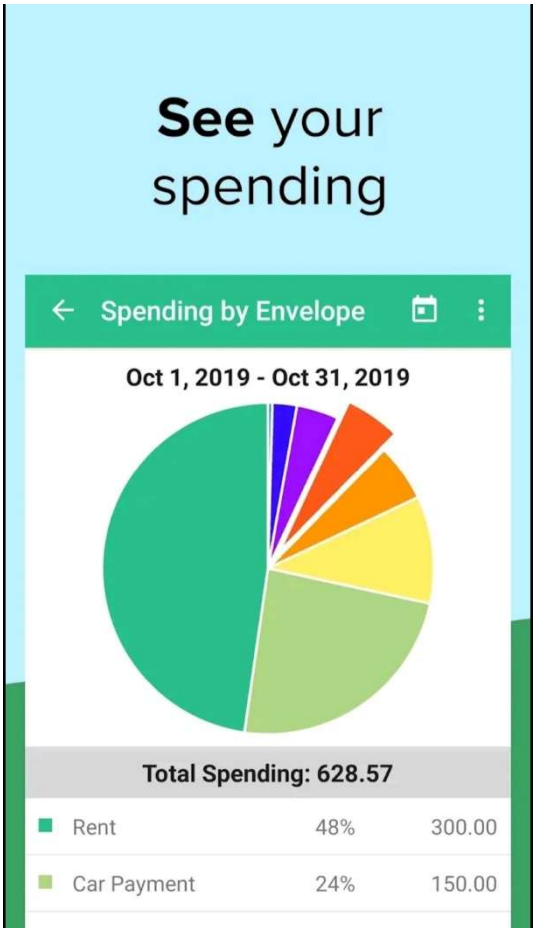


Figure 1: Goodbudget's budgeting interface, where users can see and handle their envelope distributions. It also emphasizes the app's capability to synchronize data across various devices, enabling users like couples or families to collaboratively share and oversee a combined budget in real time. This functionality aids in joint financial planning and enhances responsibility.

Figure 2: isplay of expenditures in various categories utilizing a pie chart. Every segment signifies a particular expense category like housing, transportation, or food items. This feature enables users to swiftly grasp their spending patterns and pinpoint the sectors that take up the biggest share of their budget.



Practice conscious spending

← Add Transaction ✓ ⋮

Payee

Verizon

Amount

100.00

Expense

Envelope

Cell Phone [100.00 left]

Account

My Account [853.87]

Date

10/14/2019

Check #

(Optional)

☒ Schedule this...

Every Month

Figure 3: Feature for manual transaction input in Goodbudget. Users provide information like the payee, sum, category (envelope), and date. This manual method promotes mindful spending, since users need to log every expense actively, enhancing awareness and oversight of their financial behaviors

Figure 4: Demonstrates how Goodbudget monitors expenditures within a certain category. It indicates the leftover balance and if the user is adhering to the budget. The interface shows recent transactions, enabling users to track their financial activities and confirm they remain within their designated limits

Stay on budget

← Cell Phone 🔍 ⋮

Cell Phone

100.00

100.00

Great! You're ahead by 90.32.

Transactions

10/14 [Pending] Verizon

100.00

⊕ My Account

-0.00

10/01 Income

+100.00

My Account

100.00

+

Figure 5 & 6: Demonstrates Goodbudget's envelope distribution and budgeting planning functionalities. The initial screen displays how users distribute their income into various spending categories (envelopes) like groceries and savings prior to making any purchases, encouraging proactive financial management. The second screen shows the editing and management of these envelopes, allowing users to set budget limits for each category and track remaining balances. Collectively, these attributes facilitate organized budgeting by motivating users to anticipate their expenditures ahead of time and retain oversight of their financial assets

Create a budget that works

Edit Envelopes

ADD ENVELOPE **GET UNLIMITED ENVELOPES**

Monthly (8) 2 of 10 free Envelopes left

Groceries	150
Gas	65
Eating Out	50
Giving	100

More Envelopes (4) 6 of 10 free Envelopes left

Annual

Savings	8
---------	---

Estimated **Monthly** Income **1,200** **Remaining -33**

Plan before you spend

Fill Envelopes

FROM NEW INCOME **FROM UNALLOCATED**

Fill from New Income

Received from
1st Paycheck

How to fill Envelopes
Fill Each Envelope

Amount
1200.00

Account
My Account

Date
10/01/2019

☐ Schedule this...

Your Envelopes

Groceries	Add budgeted amount of 150.00
-----------	-------------------------------

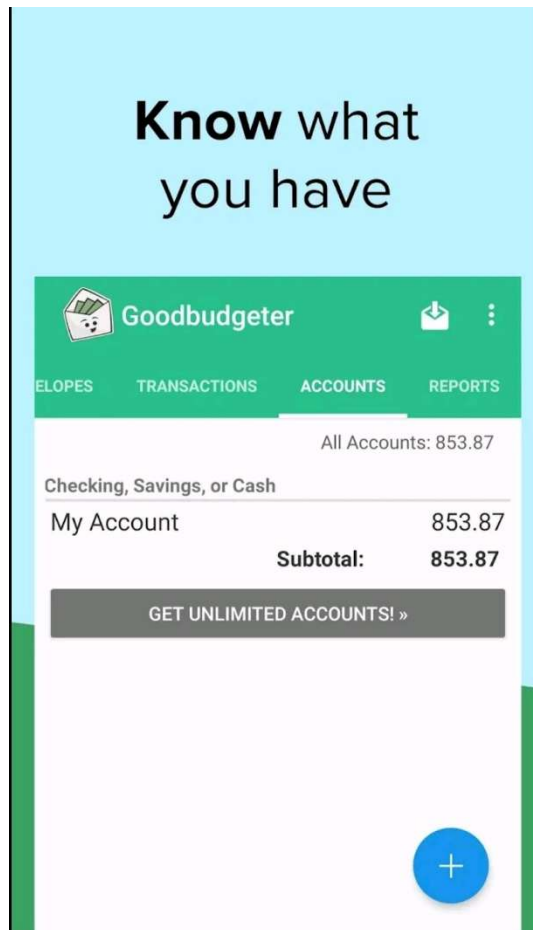
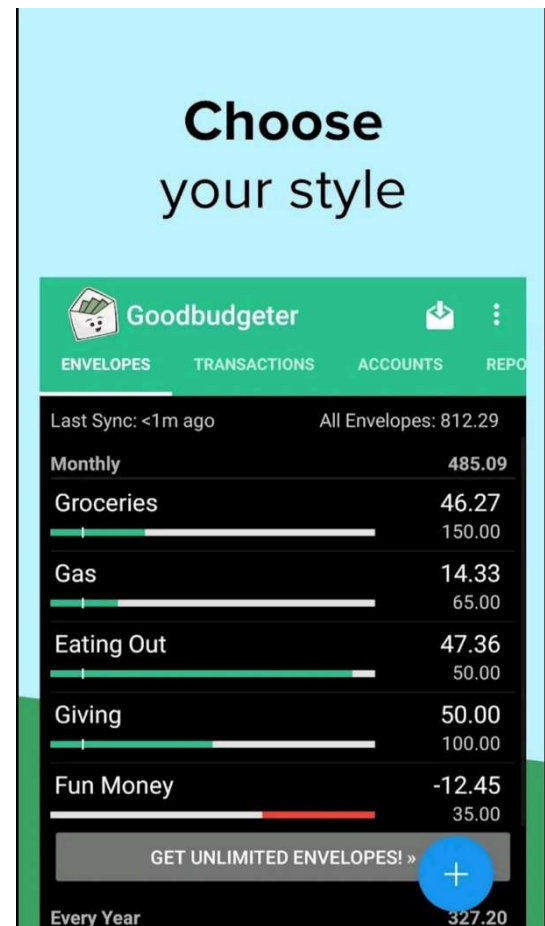


Figure 7 & 8: Goodbudget's interface customization options and account overview features. The initial screen displays the accounts area, allowing users to see their overall available balance across different accounts like cash, savings, or checking. This gives users a distinct insight into their complete financial status. The second screen showcases the app's interface customization, featuring a dark mode option that improves user experience and accessibility. It also shows envelope balances and expenditure progress, enabling users to visually monitor their budget while customizing the interface to suit their individual preferences.

Strengths

- **Effective budgeting discipline:** Manual entry forces users to be aware of their spending (Clements, 2022).
- **Cross-platform syncing:** Ideal for couples or families sharing financial responsibilities.
- **Simple interface:** Low learning curve and minimal clutter.
- **Reliable offline functionality:** Users can work without constant internet access.



Weaknesses

- **No automatic bank syncing:** Users must manually enter every expense, which becomes time-consuming (Perez, 2021).
- **Limited analytics:** Reports are basic and lack predictive or AI-based insights.
- **Not ideal for users with high transaction volume:** Daily spenders may become fatigued by repeated manual entry.

Innovative Features

- **Digital envelope budgeting system**, historically proven to reduce overspending (Thompson, 2020)
- **Household sharing**, allowing collaborative budgeting between partners, families, or roommates.
- **Cross-platform cloud sync**, keeping multiple devices updated with one budget.

2.3 PocketGuard

Overview

PocketGuard is a personal finance app created to assist users in monitoring their expenditures, controlling costs, and reaching savings objectives through a straightforward and automated method. The application is accessible on Android and aims to simplify budgeting by offering real-time financial information. In contrast to manual budgeting applications like Goodbudget, PocketGuard links to users' bank accounts and automatically classifies transactions (NerdWallet, 2023).

In a South African setting, where numerous students and young professionals handle restricted monthly finances, PocketGuard can help users monitor routine expenditures like groceries, transportation (e.g., taxis or fuel), and airtime/data acquisitions. A major feature of the app is the "In My Pocket" function, which determines how much money a user has remaining to spend after considering bills, savings, and necessary expenses.

The application features a simple and uncluttered design, allowing even novices to navigate it effortlessly. It additionally offers functionalities like tracking bills, monitoring subscriptions, and providing visual overviews of expenses. These attributes assist users in comprehending their financial habits and promote improved money management practices.

Screenshots with Descriptions

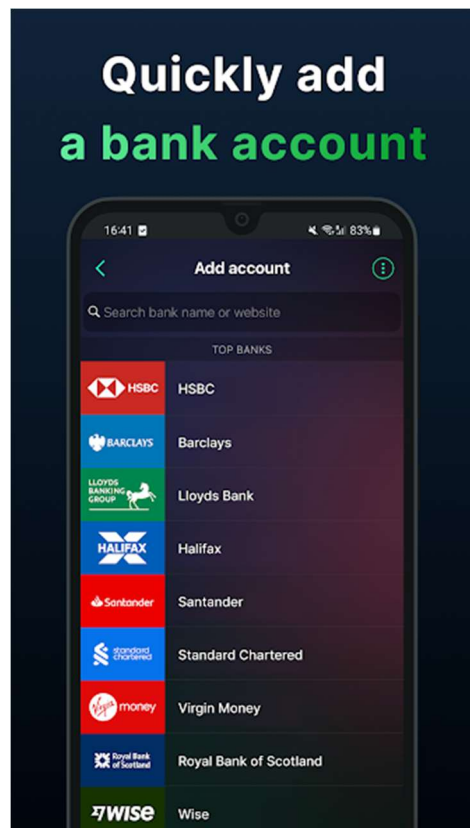
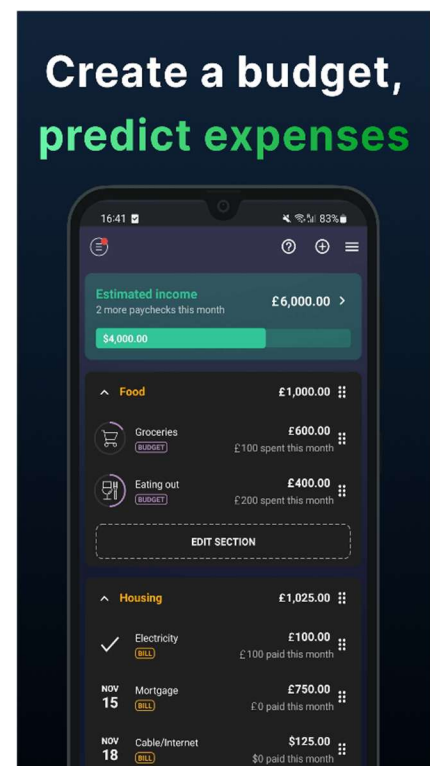


Figure 1: demonstrates the procedure of connecting a bank account to the application. It displays a roster of supported banks, enabling users to link their accounts for automatic tracking of transactions. Many of these banks are international, potentially restricting usability for South African users and necessitating manual data entry instead.

Figure 2: displays the budgeting function, allowing users to specify projected income and distribute budgets across various categories like food and housing. The system enables users to monitor expenses against established limits, assisting them in adhering to their budget and managing their finances more efficiently



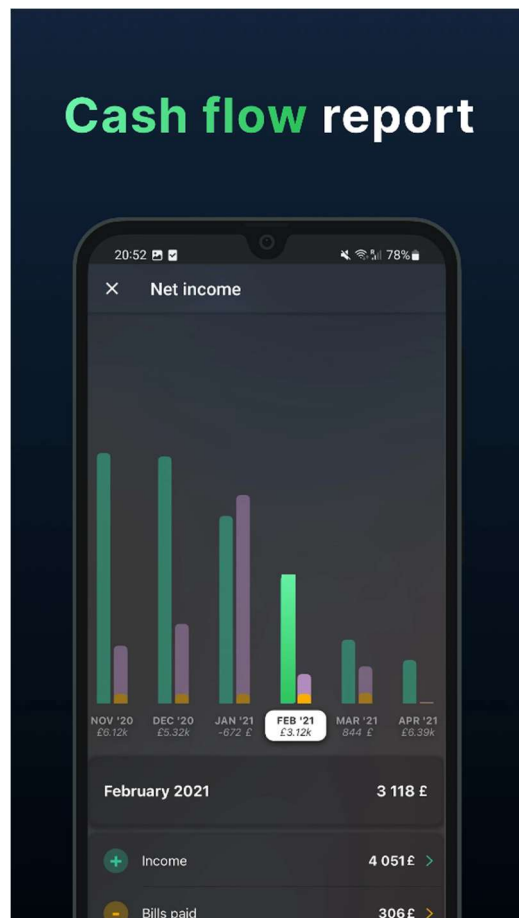
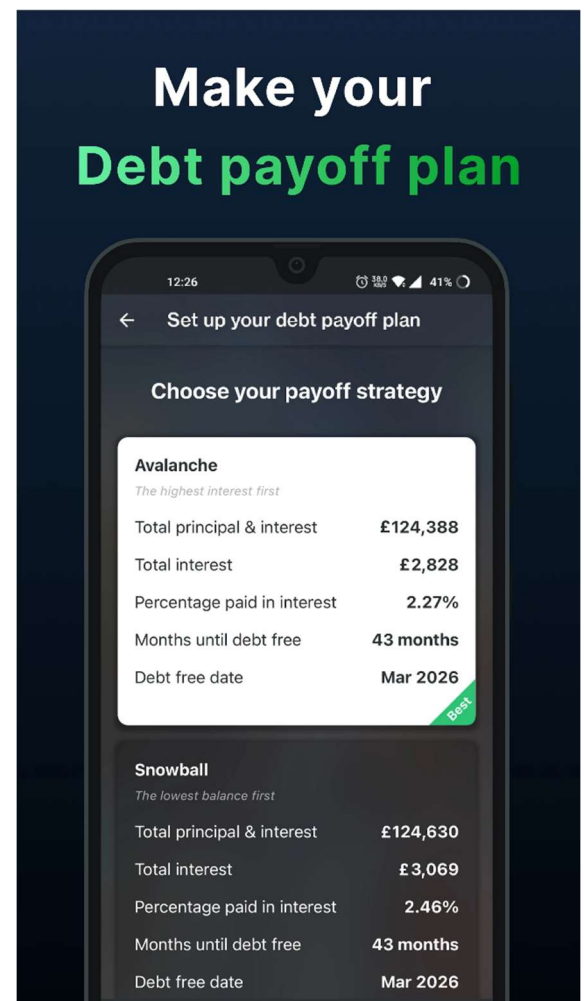


Figure 3: display offers a comprehensive analysis of revenue and costs for a particular month. It emphasizes important financial elements like overall earnings and expenses settled. This tool aids users in comprehending precisely where their funds are allocated and promotes better budgeting management.

Figure 4: Users can choose various strategies, like the avalanche or snowball technique, to handle their debt. The application offers comprehensive details such as overall interest, length of repayment, and anticipated debt-free date, aiding users in efficiently organizing and handling their financial responsibilities



Strengths

Automated tracking of expenses: PocketGuard automatically sorts transactions, minimizing the necessity for manual input and simplifying the process for users to monitor their expenses (NerdWallet, 2023).

User-friendly design: The straightforward and uncluttered design makes the app approachable for students and those who might be inexperienced with budgeting.

“In My Pocket” option: This feature allows users to easily identify how much money they can securely spend, aiding in the management of the limited monthly budgets typical in South Africa.

Updates on financial information in real-time: Individuals can track their finances each day, assisting them in preventing overspending and remaining within their budget constraints.

Weaknesses

Restricted assistance for South African banks: PocketGuard mainly caters to international banks, possibly restricting automatic syncing for users in South Africa. This might necessitate the manual input of transactions.

Premium functionalities necessitate payment: Certain advanced functionalities are exclusively found in the paid version, which might be out of reach for some users (Google Play Store, 2024).

Fundamental reporting capabilities: The app offers less comprehensive financial analysis in comparison to more sophisticated budgeting tools.

Innovative Features

"In My Pocket" budgeting method: This functionality automatically determines the remaining spending amount after expenses, making budgeting easier for users.

Automatic categorisation of expenses: Transactions are organized into categories like groceries, transportation, and entertainment, assisting users in recognizing spending trends.

Tracking bills and subscriptions: The app detects regular expenses, assisting South African users in managing monthly payments such as subscriptions, rent, or mobile data.

Summaries of expenditures: PocketGuard offers straightforward charts and summaries that facilitate the comprehension of financial trends over time

3. Infographic Comparison of Apps



4. Best Features to Include in Proposed App

Based on the analysis of the three applications, the following features are recommended for inclusion in a new budgeting app:

- Gamification elements (rewards and levels) to improve user engagement and consistency
- Envelope-based budgeting for structured financial planning and discipline
- Automatic expense tracking to reduce manual effort
- Subscription tracking and management to minimise unnecessary expenses
- Goal-setting and progress tracking to encourage long-term financial planning
- A user-friendly and visually appealing interface to enhance usability

5. Conclusion

In conclusion, personal budgeting applications play a significant role in helping individuals manage their finances effectively. The three applications analysed in this report WalletCorner, Goodbudget, and PocketGuard demonstrate different approaches to budgeting, each with its own strengths and limitations. WalletCorner introduces gamification to increase user engagement and motivation, making it particularly effective for encouraging consistent financial behaviour. Goodbudget, on the other hand, provides a structured and disciplined approach through the envelope budgeting system, which is ideal for users who prefer manual control and clear financial boundaries. PocketGuard focuses on automation and convenience, offering tools that simplify expense tracking and subscription management.

Although each application has unique advantages, no single app fully addresses all user needs. Therefore, the most effective budgeting solution would combine the engaging elements of gamification, the structure of envelope budgeting, and the efficiency of automation. By integrating these features, a new application could provide a comprehensive and user-friendly financial management tool that caters to a wide range of users.

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